Laser physics – light in step

When Theodore Maiman switched on the first laser in May 1960, some called it “a solution looking for a problem”. Today lasers are found in barcode scanners, fibre-optic cables, eye surgery, welding robots and gravitational-wave detectors. The laser is built on an idea Einstein had in 1916: light can make an excited atom emit a photon that is an exact copy of the one that struck it. This page explains how to make that happen on a large scale – with two interactive figures.

Einstein’s A and B coefficients

In 1916–17 Einstein re-derived Planck’s radiation law in a new way. He considered two-level atoms in equilibrium with radiation and found that there must be three processes:

Absorption

An atom in the ground state absorbs a photon and is excited. The rate is $B_{12}\,u(\nu)\,N_1$.

Spontaneous emission

An excited atom decays of its own accord and emits a photon in a random direction. The rate is $A_{21} N_2$.

Stimulated emission

A photon makes an excited atom emit a new photon with the same frequency, direction, phase and polarisation. The rate is $B_{21}\,u(\nu)\,N_2$.

The requirement that Planck’s law hold at all temperatures gives the relations

$$g_1 B_{12} = g_2 B_{21}, \qquad \frac{A_{21}}{B_{21}} = \frac{8\pi h\nu^3}{c^3}$$

Stimulated emission is the key to the laser: it amplifies the light, and the new photons are *coherent* with the old ones. But the second formula also shows why lasers are harder to make at

short wavelengths: the competing spontaneous emission grows as ν^3.

Population inversion

In thermal equilibrium there are always more atoms in the lower level than in the upper, distributed according to the Boltzmann factor $N_2/N_1 = e^{-h\nu/k_BT}$. For visible light at room temperature the ratio is about 10^{-35}. Light passing through such a medium is absorbed more than it is amplified. To get amplification one must create a *population inversion*, $N_2 > N_1$ – a state corresponding to a negative temperature. This is impossible with only two levels, because the harder one pumps, the more stimulated emission brings the atoms down again. The solution is three or four levels:

Three-level laser (ruby)

The atoms are pumped from the ground state to a high band, decay quickly to a long-lived level, and the laser transition goes back to the ground state. More than half the atoms must be pumped up – which requires very strong pumping.

Four-level laser (Nd:YAG, HeNe)

The laser transition ends in a level that empties quickly. The lower laser level is almost empty, so even a small population in the upper level gives inversion. Almost all practical lasers are four-level systems.

From maser to laser

The idea was first realised with microwaves. In 1954 at Columbia University Charles Townes, James Gordon and Herbert Zeiger built a *maser* (Microwave Amplification by Stimulated Emission of Radiation), in which excited ammonia molecules were separated from unexcited ones by an electric field. Nikolay Basov and Aleksandr Prokhorov in Moscow independently developed the theory. The three shared the Nobel Prize in 1964.

In 1958 Townes and his brother-in-law Arthur Schawlow showed that the principle could be extended to visible light if the amplifying medium is placed between two mirrors. Gordon Gould, a PhD student at Columbia, had the same idea, wrote it down in a notebook which he had notarised in November 1957 – and coined the word LASER. He fought for his patents for 30 years and finally won.

On 16 May 1960 Theodore Maiman at Hughes Research Laboratories in California got the first laser to work: a small ruby rod, surrounded by a helical flash lamp, gave pulses of red light at

694 nm. Several large laboratories had assumed that ruby could not be used. In December of the same year Ali Javan, William Bennett and Donald Herriott at Bell Labs built the first gas laser, helium–neon, and in 1962 the first semiconductor lasers appeared. See the laser animation.

Threshold and rate equations

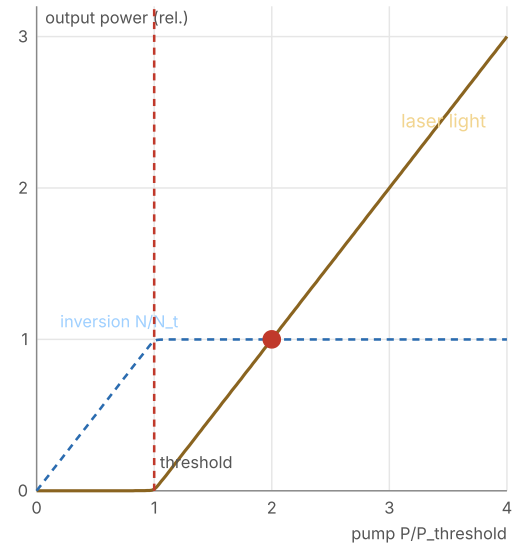
A laser oscillates when the gain on a round trip in the resonator exceeds the losses – through the partially transmitting output mirror, by scattering and by absorption. For a gain medium of length L, gain coefficient g and mirror reflectivities R_1 and R_2, the threshold condition is

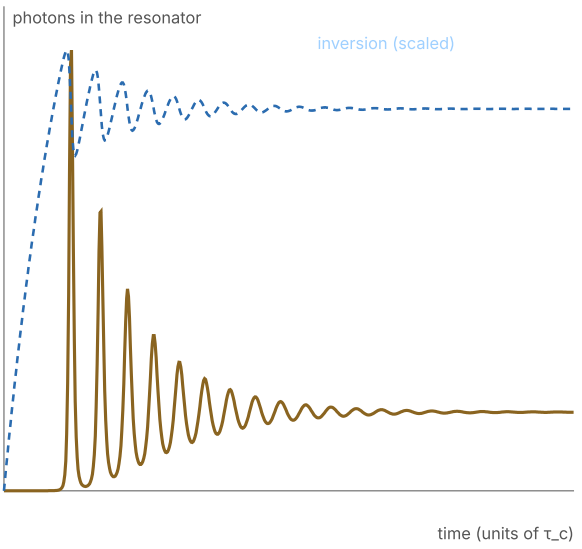
$$R_1 R_2\, e^{2(g-\alpha)L} = 1 \quad \Rightarrow \quad g_{\text{threshold}} = \alpha + \frac{1}{2L}\ln\frac{1}{R_1 R_2}$$

The dynamics can be described by two coupled rate equations for the inversion N and the number of photons ϕ in the resonator:

$$\frac{dN}{dt} = P - \frac{N}{\tau} - BN\phi, \qquad \frac{d\phi}{dt} = BN\phi - \frac{\phi}{\tau_c} + \beta\frac{N}{\tau}$$

Here P is the pump rate, τ the lifetime of the upper level, τ_c the lifetime of photons in the resonator and β the small fraction of spontaneous emission that goes into the laser mode. Below threshold the inversion grows with pumping, and only weak spontaneous light comes out. Above threshold the inversion is “clamped” at the threshold value, and all extra pump power becomes laser light. When the laser is switched on, the output typically oscillates for a while like a damped spring – *relaxation oscillations* – before settling down.



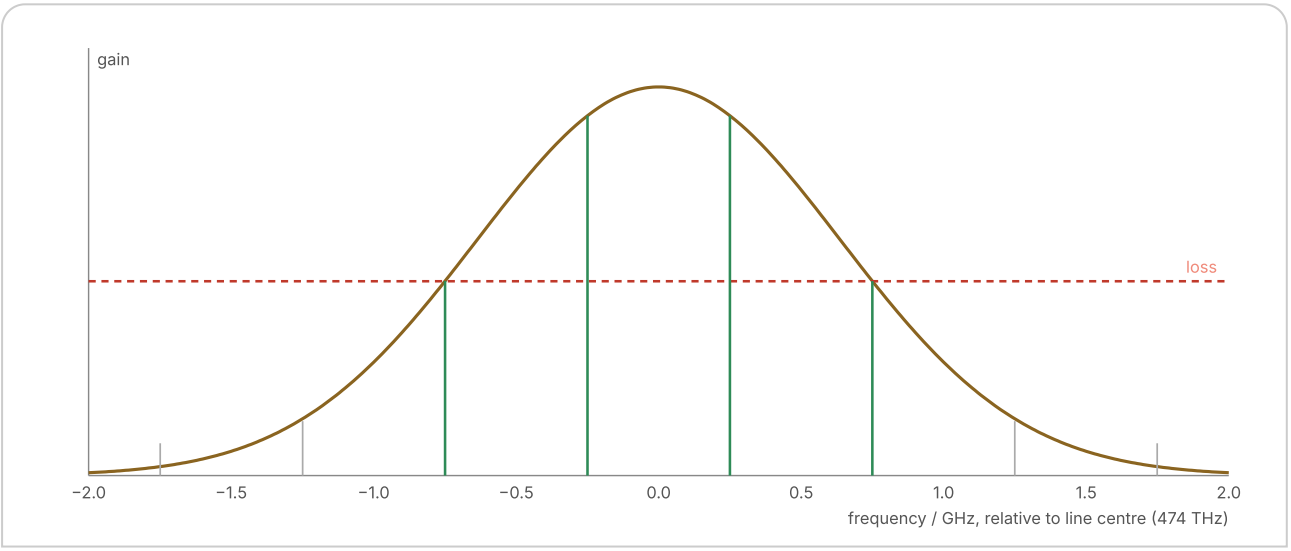


The resonator and longitudinal modes

Between two mirrors a distance L apart, only standing waves with a whole number of half-wavelengths can exist, $L = q\lambda/2$. The frequencies are spaced by

$$\Delta\nu = \frac{c}{2L}$$

For a 30 cm HeNe laser this is 500 MHz. The gain medium amplifies only within a certain frequency band – for HeNe about 1.5 GHz wide because of the Doppler effect from the motion of the gas atoms. The laser oscillates on all the modes within the band where the gain exceeds the losses. A short laser can therefore oscillate on a single frequency, a long one on many. With an extra etalon in the resonator one can select a single mode and get a linewidth below 1 kHz – a relative precision of 10^{-12}.



Gaussian beams

The beam from a good resonator has a cross-section that is a Gaussian curve. It is narrowest at the *waist*, with radius w_0, and then spreads out:

$$w(z) = w_0\sqrt{1 + \left(\frac{z}{z_R}\right)^2}, \qquad z_R = \frac{\pi w_0^2}{\lambda}, \qquad \theta \approx \frac{\lambda}{\pi w_0}$$

Within the Rayleigh length z_R the beam is almost parallel. A HeNe beam with a 0.5 mm waist has a divergence of only 0.4 milliradians. Once the Apollo astronauts had placed reflectors on the Moon in 1969, laser pulses were sent up through a telescope: on the Moon the spot was a few kilometres across, and today the distance is measured with millimetre precision (see The Moon's

orbit in 3D). Laser light can be focused to a spot the size of the wavelength – which gives the enormous intensities.

Types of laser

Laser	Wavelength	Medium	Typical use
Ruby (1960)	694 nm	Cr^{3+} in Al_2O_3	historical, holography
Helium–neon (1960)	633 nm	gas discharge	alignment, interferometry
CO_2 (1964)	10.6 µm	molecular vibrations	cutting and welding (kW)
Nd:YAG (1964)	1064 nm	Nd^{3+} in a crystal	industry, medicine, frequency-doubled to 532 nm
Semiconductor (1962)	400–2000 nm	p–n junction	fibre optics, barcodes, Blu-ray, laser pointers
Ti:sapphire (1982)	650–1100 nm	Ti^{3+} in Al_2O_3	femtosecond pulses
Fibre laser	1–2 µm	doped glass fibre	industry, up to 100 kW
Free-electron laser	X-rays to microwaves	electron beam in an undulator	structure determination (e.g. European XFEL)

Short and powerful pulses

In *Q-switching* the resonator is kept closed while the inversion builds up and then opened suddenly: all the energy comes out in a pulse lasting nanoseconds. In *mode locking* many longitudinal modes are forced to oscillate with fixed phases relative to one another; they interfere to form a train of pulses that can be shorter than 10 femtoseconds – a few cycles of light. The pulse duration is inversely proportional to the bandwidth, $\Delta t\,\Delta\nu \gtrsim 0.44$.

In 1985 Gérard Mourou and Donna Strickland developed *chirped pulse amplification*: a short pulse is stretched in time, amplified without destroying the amplifier and compressed again. It gave intensities above 10^{22} W/cm² and the Nobel Prize in 2018 – the same year Arthur Ashkin received it for the optical tweezers that can hold living cells with light. Pierre Agostini, Ferenc Krausz and Anne L'Huillier received the Nobel Prize in 2023 for attosecond pulses that can photograph the motion of electrons.

Applications

Communication

Semiconductor lasers send data through glass fibres at terabits per second. In 1966 Charles Kao showed that ultrapure glass could carry light over long distances (Nobel Prize 2009).

Measurement

LIGO measures gravitational waves with laser interferometers that register changes of a thousandth of the diameter of a proton. Optical atomic clocks are so precise that they would be out by less than a second over the age of the universe. See Relativity.

Laser cooling

Light can slow atoms down to microkelvin temperatures (Chu, Cohen-Tannoudji and Phillips, Nobel Prize 1997) – the road to Bose–Einstein condensates and atomic clocks.

Medicine and industry

Eye surgery, tattoo removal, welding, cutting, 3D printing of metal and the manufacture of computer chips with extreme ultraviolet light at 13.5 nm.

Further reading

- Orazio Svelto: *Principles of Lasers* (5th ed., Springer 2010)
- Anthony E. Siegman: *Lasers* (University Science Books 1986)
- Jeff Hecht: *Beam: The Race to Make the Laser* (Oxford 2005)
- Charles H. Townes: *How the Laser Happened* (Oxford 1999)

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